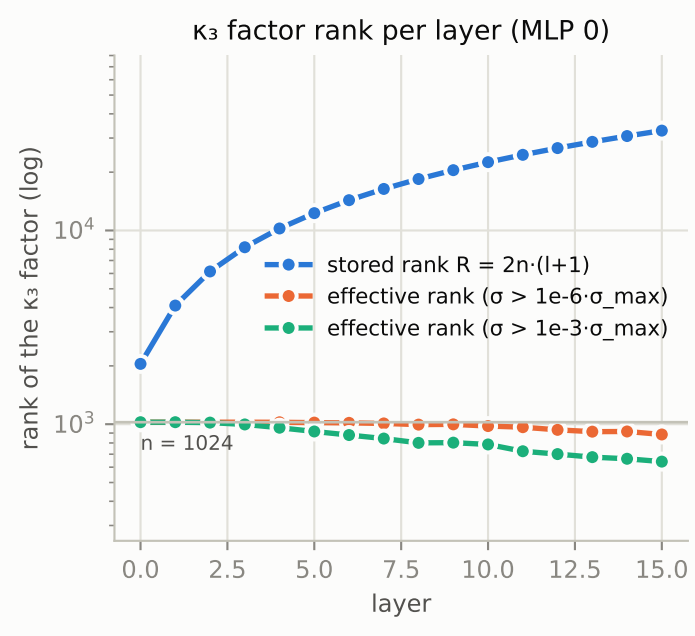
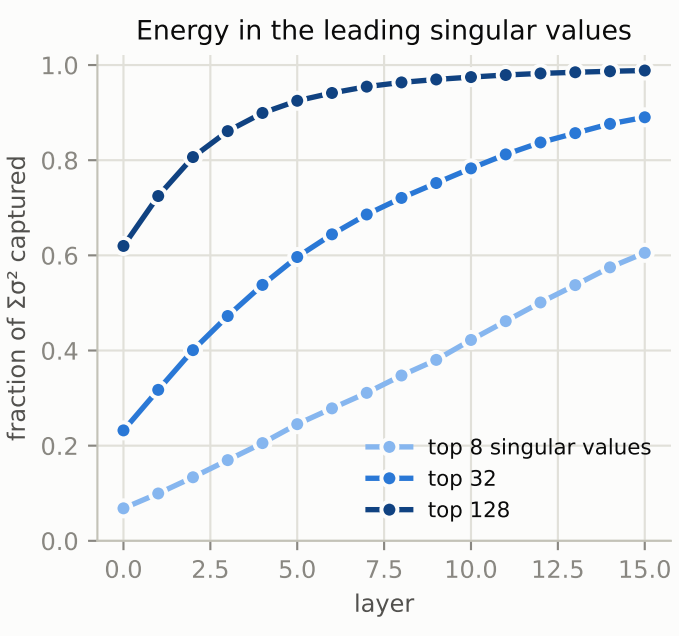
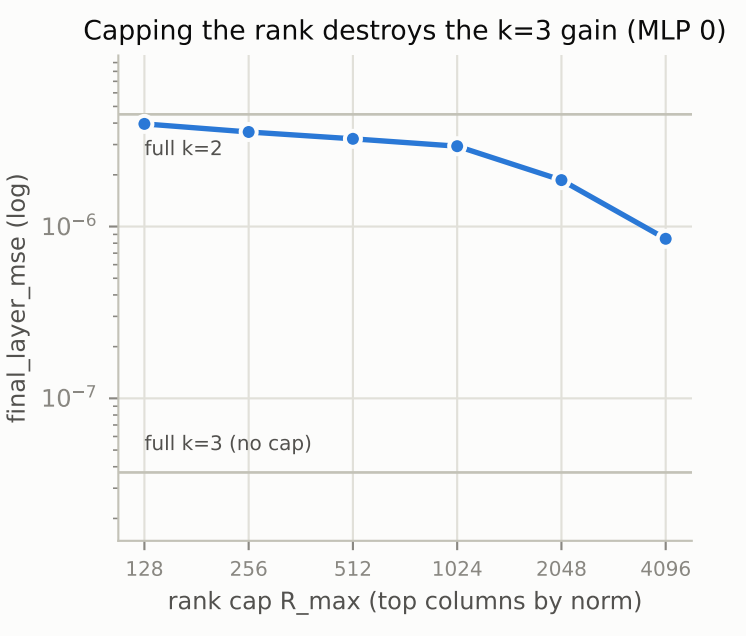

κ₃ factor rank per layer (MLP 0)
rank of the κ₃ factor (log)
10^4
10^3
stored rank R = 2n·(l+1)
effective rank (σ > 1e-6·σ_max)
effective rank (σ > 1e-3·σ_max)
n = 1024
0.0
2.5
5.0
7.5
10.0
12.5
15.0
layer


Energy in the leading singular values
fraction of Σσ² captured
1.0
0.8
0.6
0.4
0.2
0.0
top 8 singular values
top 32
top 128
0.0
2.5
5.0
7.5
10.0
12.5
15.0
layer


Capping the rank destroys the k=3 gain (MLP 0)
final_layer_mse (log)
10^{-6}
10^{-7}
full k=2
full k=3 (no cap)
128
256
512
1024
2048
4096
rank cap R_max (top columns by norm)